“Twins (Not) Separated at Birth: Genetic Effects on Partisan Identities”

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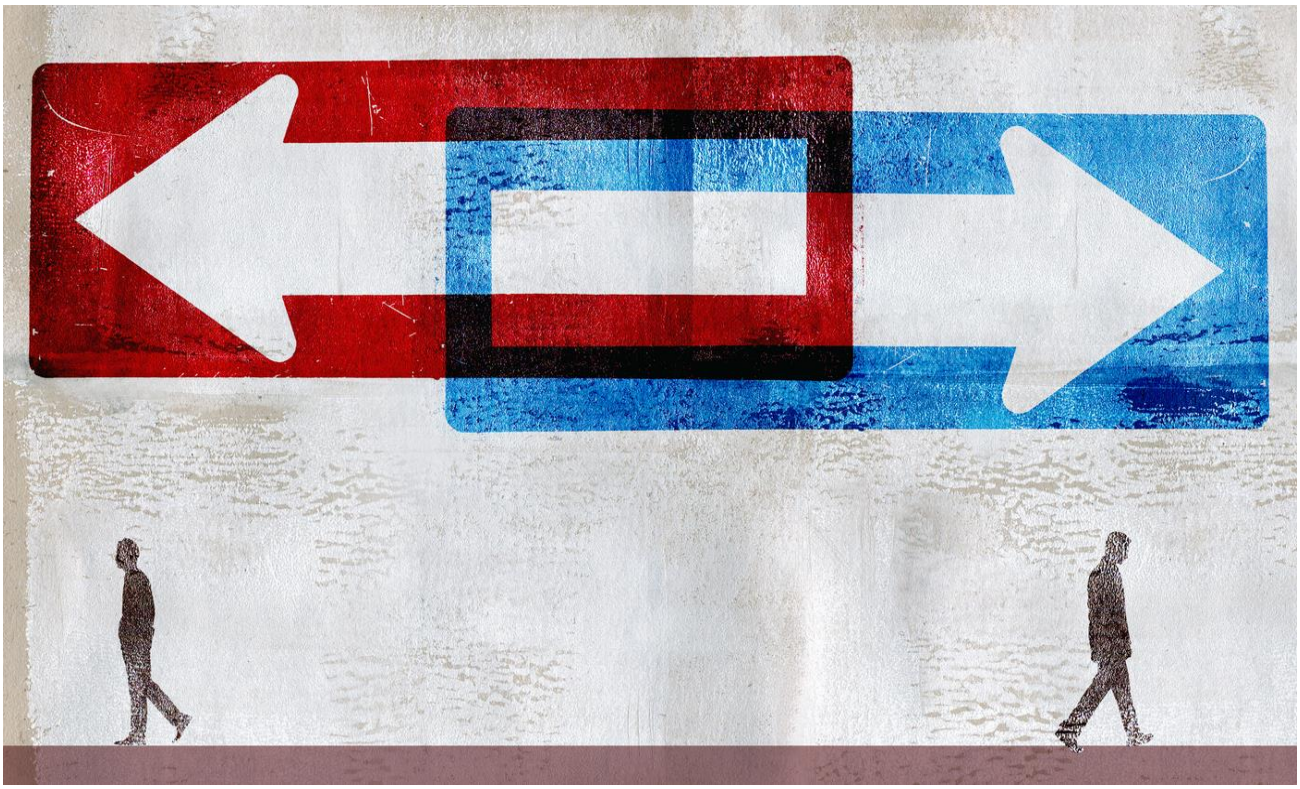

The classic storyline of twins separated at birth is simply fascinating to the public. If one were to google it, then one would find numerous incidents throughout the globe even though it represents an extremely rare occurrence in daily life. Our concerted interest in the topic exists largely because it raises important questions about the origins of human behavior. Specifically, it provides valuable leverage about the role of *nature versus nurture* in human development. We can use the isolation of genetically similar twins to better understand how social context influences the formation of individual beliefs and identities. Twins-separated-at-birth perspective allows us to think about what factors shape our life stories, if we are predestined to certain paths by biological attributes, and whether real events can disrupt that pathway.

Take, for instance, a hypothetical scenario of a pair of twins born into difficult circumstances. They have parents who live in poverty and who constantly struggle to make ends meet. Two extra mouths to feed puts them past the point of sustainability and the parents are forced to make a difficult decision. Because they are unable to provide effective care for their children, they are forced to unthinkable ends. The adoption mechanism is their only resource. They simply must give up their children for them to survive. Over the following years, the siblings grew up apart from each other. They transition into adulthood under unique contextual circumstances. Their development and maturation are products of entirely different environments.

Assume, for instance, our first twin was adopted by a Houston family who lives in an urban middle-class neighborhood. Both parents are high school graduates. The father has a decent-scale local manufacturing plant. The mother works as an administrator in a small business office. work long hours to make ends meet and do not have much time

for church attendance. The family exists in an environment of potential economic decline. The threat to their economic well-being comes from an ever-present plant shutdown or it is a function of injuries that render the father unable to work. This twin is going to have more limited options in terms of education and the prevailing threat of economic decline may be critical to future life choices.

Alternatively, assume the second twin is adopted by a well-off family in El Paso. The family wealth is tied to significant land holdings with oil deposits and decent grassland for cattle ranching. The father had an advanced science degree from the University of Texas Austin. The mother graduated with a law degree from Baylor University. They are deeply tied to their local Baptist church and are considered church leaders. This twin will be born in a financially secure environment with many available options for the future. This sibling will have more educational opportunities, can engage in more extensive travel, and will grow up in a family context with a balance of work and leisure. This financial security will contribute to a broader array of life choices and these twins may be quite different as they approach adulthood.

If we asked you to predict the outcome of the above twin story, we would have clear ideas of how these twins might develop into mature politically active adults. We would expect our first twin to be more inclined to support the Democratic Party. The second twin, on the other hand, should be more likely to support the Republican Party. This first twin is raised in a blue urban environment under persistent economic threats and should grow to value the social safety net policies targeting working-class voters. The second is raised in the ruby red rural environment and has ample financial security that may lead to support for business and strong social conservative values. On the surface, we would

expect these twins to have different political identities, but what if they both turn out to be Democrats or Republicans? Our focus on economic threat and surplus is not as critical to the partisan calculus as we might think. Political scientists love to think about this type of scenario because it helps us learn about where our political preferences and identities come from.

Traditionally, political scientists point to a process of *political socialization* (Campbell, Converse, Miller, and Stokes 1960) as the means through which we develop our political identities. This theory implies that we learn our political values like other types of knowledge. For example, in the above scenario, our twins are born into unique regions, states, and local economic contexts. The siblings were raised by their families to value certain principles. They gain knowledge in diverse types and levels of educational institutions. They may also have quite diverse types of peers and unique individual interests versus each other. Finally, they may be affected quite differently by critical world events and combat potential. Political socialization would imply that all these factors combine to influence how our twins will develop their partisan identities.

The alternative to political socialization can be found in the new research paradigm of *biopolitics* (Somit and Peterson 1998). The goal of biopolitics is to increase our knowledge of political systems by tapping into the methods and knowledge found in the hard sciences like chemistry and biology. It argues that humans might be genetically predisposed to engage in certain political behaviors. The research incorporates new genetic sequencing technologies that can pinpoint certain genomes that might affect our political preferences and identities. It uses technology (e.g., medical imaging) to better

measure political variables. However, it has a concerted focus on the balance of contextual influences versus effects related to the human genome. In this way, our hypothetical twin story is relevant. Biopolitics scholars want to know if genes drive the formation of partisan identities versus social context. Twin studies are extremely valuable in this respect as they offer us valuable leverage on genetic effects versus social context.

Our project aims to better understand the relationship between genes and political party identities. We employ a classic twin research design to evaluate relationships between genes and political identification. The study comprises a sample of twins that vary in their genetic overlap. We include a subsample of homogenous identical twins with the same genetic makeup. We then compare it to a subsample of heterogeneous fraternal twins. In this case, we take advantage of these pairs of twins being not separated at birth. Our study captures twins that are exposed to similar processes of political socialization. As a result, we hope to isolate partisan differences associated with genetic differences. In the end, we seek to answer the following research question: Are differences in genetic overlap found between diverse types of twins associated with differences in political party identification?

Exploring the intersection of twin studies and biopolitics

To answer our research question, we will discuss four pieces of political science literature. The first collection of research considers the development of political preferences and the prevalence of rational voters within the public. We will then take up the work on political party identification and cover the process of political socialization. After that, we will introduce the new biopolitics paradigm and cover its origin story. Finally,

we will cover the class twin research design and our knowledge of genetic effects on political preferences and partisan identities. With those four kinds of literature in hand, we will present our working theory for genetic influence and state testable hypotheses to answer our research question.

Political Preference Research

A political party is an organization that works toward the goal of winning elections, creating laws, and running the government. The Democratic Party and the Republican Party are the two main political parties in the United States. Political parties are organized organizations that collaborate to obtain political power and then utilize that power to affect policy. Parties compete with one another to win elections and remain in power.

To evaluate how ideology is conceptualized, Converse (1964) examines responses to open-ended interview questions. He concludes that the liberal-conservative continuum is an example of a high-level abstraction that the average person does not often employ because of the volatility of responses and the lack of links created between replies. Most individuals do not have a coherent set of core beliefs; rather, they have a variety of unrelated viewpoints. Large segments of the voters do not have views that are consistent with one another, even on matters that are extremely contentious and widely discussed. Many individuals respond to survey questions as if they are tossing a coin.

The competing explanation was proposed by Achen (1975), who looked at Converse's results from a statistical perspective. He contends that people have unstable political beliefs and attitudes that appear to oscillate not because they do not have any political opinions or because their opinions are only tangentially related to one another, but rather because of the nature of the survey questions that are asked of them. Achen constructed

a model that was intended to split ideological restriction into two separate causes, namely the instability of voters' opinions and the poor reliability of survey questions so that he could evaluate the hypothesis that this was the case. Achen wanted to demonstrate that the volatility of public sentiments was mostly because survey questions tend to be ambiguous and subject to change. Achen discovered higher levels of consistency amongst individuals' perspectives after making the necessary adjustments to the study to account for this inaccuracy. In addition to this, if this measurement inaccuracy is accounted for, the amount of attitude restriction that exists also rises.

The writers of this book on electoral behavior, which won an award and became a great seller, have brought their research up to date by doing a detailed examination of the presidential election that took place in 1976. Again, by meticulously studying national voting trends, they offer substantial data meaning. Professors Norman Nie, Sidney Verba, and John Petrocik show how the campaign's political backdrop seemed to affect the significance that issues seemed to play in the 1976 voting process. In this regard, the 1976 elections were a repeat of prior fights like those in 1960. In some respects, such as the weakening of partisanship, the election of 1976 was comparable to elections held more recently. The authors put their past research on the electorate to the test by using the 1976 election as a case study. This book demonstrates that the average American voter is not a "new" voter but rather a changeable one. The 1976 election is an example of the changing character of the public's conduct and sentiments.

Alan I. Abramowitz (2010), a renowned political scientist, makes a ground-breaking argument in a new book that the most significant divide in American politics is not between the left and the right, but rather between politically engaged citizens and those who are

not. He contends that this divide is the most significant. According to his theory, the members of the public who actively participate in political discourse are the ones who most faithfully represent democratic citizenship values; nevertheless, this is also the demographic that is the most politically divided. So, polarization at the highest levels of government is not an indication that elites have disconnected themselves from the public; rather, it reflects their response to the most politically involved portions of that population. Polarization is often seen to be harmful to democracies, but Abramowitz concludes that it may assist in promoting electoral accountability and boost public interest and engagement in politics by providing voters with clear alternatives.

The study that Zaller (1992) conducted gives support for the Converse premise, which shows that individuals do not have fixed opinions and are more likely to provide fluid, off-the-cuff replies to questions that are posed to them. Zaller suggested that most individuals do not have a logical framework for their political ideas and that they instead depend on "considerations" that spring to mind when questioned about a specific problem. The study of Zaller casts doubt on the hypothesis that individuals can maintain coherent and consistent perspectives about political matters. Instead, he thinks that people's ideas are far more flexible and dependent on the environment in which they are expressed. This has significant repercussions for the methods used to assess and evaluate public opinion and for the ways political campaigns are run.

The Role of Party Identification

Party identification is a key idea in political science because it sheds light on the mechanisms that influence people's political worldviews. It refers to a person's emotional commitment to a political party, which is frequently created via socialization, and

subjective experiences. Voters use party identification as a hint when choosing candidates and evaluating issue opinions because it gives them a quick way to gauge a candidate's morals and ideologies. The idea of party identification is crucial to political science and aids in our comprehension of the intricate processes that influence political behavior and decision-making. Political socialization implies that all these factors influence how our twins develop their partisan identities.

An innovative study on the connection between party identification and voting behavior was carried out by Campbell, Converse, Miller, and Stokes (1960). According to their research, party identification is the primary factor driving voting and serves three essential functions: it informs voters, boosts participation/ turnout, and affects the results of elections.

Those who identify with a certain political party are more likely to hold a set of values and ideas that are consistent with the platform of that party. Voters can more quickly understand complicated political topics because of this alignment, which also makes decision-making easier. As a result, party identification serves as a heuristic that makes it easy for voters to make informed decisions. Encouragement of involvement and turnout is the second function of party identification. According to research, those who identify with a political party are more likely to cast ballots than those who do not. Since they are more likely to consider voting as a civic obligation and feel a feeling of responsibility to engage in the political process, those who strongly connect with a party are more likely to experience this effect. (Campbell, Converse, Miller, and Stokes 1960). Influencing election results is the third task of party identification. Even if they do not agree with the candidate's exact policy stances, people who identify with a certain party are

more inclined to vote for candidates who support that party's program. This impact is especially potent in highly politicized settings because voters are more prone to cast their ballots along party lines.

Our comprehension of voting behavior has been dramatically influenced by the studies of Campbell, Converse, Miller, and Stokes (1960). Their results indicate that party identification is important in determining the political environment, giving voters informational cues, promoting involvement/ turnout, and affecting voting results.

Party identification is a learned behavior developed over time through social interactions with family, friends, and institutions. Politics and party affiliation may be impacted by parents and the nurturing atmosphere they create. Party identification can also be influenced by geographic considerations, such as the political lean of one's area. Peer pressure is another factor that can affect party identification, particularly in adolescence and the initial stages of adulthood when people are more prone to want the approval of their peers. Moreover, joining a particular group, such as a club, workers union, or religious organization, may promote socialization. Hence, major societal occurrences like conflicts, financial crises, or large political scandals can impact party identification by altering people's perceptions of the viability and competence of parties. The political socialization thesis can be used to explain how party identification develops and how it affects people's political beliefs and behavior.

Converse (1969) stated that a variety of factors, such as parents and the nurturing environment they provide, regional characteristics, peer pressure, belonging to a group, and significant societal events, determine acquired behavior. While Jennings and Markus (1984) stressed the significance of group membership in influencing identity and behavior,

Jennings, et al. (1979) further emphasized the significance of peer influence in influencing attitudes and behaviors. Jennings, Stoker, and Bowers (2009) argued that crucial public events, such as political campaigns or social movements, can also impact learned behavior by influencing attitudes and beliefs about crucial problems. These studies collectively demonstrate that learned behavior is a complicated phenomenon that is influenced by a range of circumstances and that comprehending these elements is crucial for comprehending how people and groups behave in various contexts.

In American politics, party identity has always played a significant role, influencing everything from voting patterns to policy views. Dalton (2012) asserts that this tendency might be altering. Party identification among Americans has decreased recently, with fewer people connecting themselves with political power and party memberships. According to Dalton (2012), many variables, including the increase in independent voters, are to blame for the drop in party identification. The number of people who do not identify with either major political party is rising as more Americans reject the conventional opposition between Republicans and Democrats. The decline in faith in political institutions could be one factor behind this trend. According to Dalton (2012), many Americans are disenchanted with politics and have doubts about the ability of either party to run the country efficiently. Many people have become so disillusioned that they completely reject belonging to a party and choose a more personal approach to politics. The United States' changing demographics is another factor influencing the reduction in party identification. Traditional party connections may no longer be as influential as they once were as the nation becomes more diversified and home to an increasing number of minority groups.

Despite these changes, party identity continues to play a significant role in American politics. Parties continue to be important in influencing policy and energizing the electorate, as noted by Dalton (2012). To remain relevant in the years to come, parties will need to adjust to shifting demographics and new political realities, according to the drop in party identification.

The commonly accepted view that there has been a fall in partisanship in American politics is disputed by Bartels (2000). He contends that while voters may be rejecting party names, their voting behavior is not necessarily becoming less politicized. Conversely, more independent learners are now voting regularly for one party despite not accepting the name of a particular party. Bartels provides proof for his claim from several sources. He observes that over time, with only small variations, party affiliation has remained steady. He also emphasizes that although the proportion of voters who identify as independents has grown, most of these voters still have a strong preference for one party over the other. In actuality, the proportion of true independents who have no leanings toward either party has stayed stable. Bartels also disputes the notion that voters are less inclined to follow party lines and are growing more ideologically diverse. He points out that the distribution of ideologies among voters has not changed much over time, and that people who do not identify with a party are just as likely to have strong ideologies as those who do. Bartels notes that some voters reject the party designation and do not consistently support one party despite these findings. But he contends that they are a small percentage and do not indicate a substantial political trend in America.

The consensus on the collapse of partisanship in American politics is contested by Bartels' work. Voters may be rejecting party affiliations, but this does not necessarily

mean that they are becoming less political in their voting choices. However, there has been an increase in the number of independent learners who, though they may not identify as members of a particular party, consistently support that party.

Despite party identification being considered the dominant influence in voting behavior, recent research suggests that socialization plays a significant role in shaping political attitudes and beliefs. However, alternative perspectives are emerging, challenging the traditional notion that party affiliation drives voter behavior.

The Rise of Bio-Politics

An innovative approach aims to advance our understanding of political affiliation, in the form of a new research paradigm of biopolitics (Somit and Peterson, 1998). Biopolitics is the study of biological influences on political behavior. The field of biopolitics incorporates hard science research methods and knowledge to correlate genetic influences on political behavior (Somit and Peterson, 1998). The paradigm is rooted in a Neo-Darwinist theory that genetics influence behavior. The use of hard science methods provides empirical data that scientifically measures political behavior.

The seminal works of Somit (1968) and Wahlke (1979) outline the biopolitical field's foundation as a means of a new research path and subsequent paradigm in the political science realm. Wahlke (1979) argues that while behavioral ideology introduced the use of empirical data, it cannot overcome its shortcomings. The ideology does not correlate human behavior with the broader political science realm. Secondly, the discipline is grounded in general behavioral theory, which is a flawed concept of political science (Wahlke,1979). Wahlke encourages political scientists to gain knowledge on how to

surpass behaviorism and looks toward biological advances in the political realm to that end.

Somit (1968) expounds on the importance of evolutionary ethology and psychopharmacology to biopolitics. Ethology is the study of evolutionary changes in animal behavior (Somit, 1968). As it relates to political science, we are concerned with how natural selection shapes behavioral patterns that are as indicative of a species as phenotypical traits. This forms the basis of how certain behaviors are hardwired into our genetic makeup. The correlation in turn can affect our political behaviors, advancing the ideology that behavioralism is rooted in biology and biopolitics. Psychopharmacology is the study of how drugs affect the human mind. Psychoactive drugs influence human behavior, effectively changing the normal physiological behavior in the mind. This knowledge helps to address the key gap in biopolitics, which is linking behavior to biological makeup. being able to discern whether a behavior is learned or is a result of a "biological predisposition" (Somit 1968, 560). If drugs can alter the "normal" state of the mind, then this leads to behaviors being linked through genetic makeup.

Political science scholars began to seek a physiological approach to behaviorist theory and their scientific method. The drawback to behavioral science is the overreliance on "verbal self-report" (Lodge and Wahlke 1972, 4). The design is inherently plagued with subjective feelings that fail to provide sound empirical data that hard science like physiology could help provide. Hirsche and Wiegele (1981) suggest that new tools, such as the voice stress test can assist in gaining more measuring tools for data.

Biopolitics is essential in opening the conversation about how physiology leads to a new focus on political behavior. Specifically on how genetic effects influence political behavior. Genetics might help to pinpoint what effects can influence our political preferences and subsequent political parties. Specifically, we want to observe how genetics affect one's political preferences and subsequent political party affiliations.

Genetic Effects on Party Identification and Ties

Eaves and Eysenck (1974) laid the groundwork for the exploration of genes and political behavior by being the first to explore monozygotic and dizygotic twins and their variance of attitudes towards a variety of ideologies. Contrary to the beliefs of life science studies preceding their work, their findings pointed toward an elevated level of correlation between monozygotic twins compared to dizygotic twins. This suggested that genetic transmission may be of higher importance than social learning. Eaves and Eysenck used a classical twin research design in their study, examining pairs of twins who were raised alongside one another. While socialization is inevitable, one can assume twins receive the same or negligible effect of socialization on their respective ideologies allowing for only variance in attitudes to be measured. Using twins allowed for this socialization factor of each studied pair to be constant. Also, twins offer genetic variance, helping understand how much genetic influence affects attitudes toward ideologies. Identical twins provide 99% genetic overlap, and fraternal twins provide 50% to 75% genetic overlap hence the differentiation between monozygotic and dizygotic twins in the study.

The classical twin design study is paramount to Eaves and Eysenck for many factors. The most important being that the study, by design, controls socialization

creating an area of consistency and allowing for the isolation of key variables. The variables examined being concordance in attitudes towards different political ideologies in twins help researchers understand the biological effects on party identification.

Eaves, Heath, Martin, and Maes (1999) was another groundbreaking moment highlighting classical twin design studies. The researchers collected close to 30,000 surveys from sets of twins evaluating the correlation between four criteria regarding personality and six involving social attitudes. Eaves, et al, successfully found evidence of genetic influence on personality and social attitudes that they published in their study. Although they did find evidence of genetic influence on personality and social attitudes, they include a caveat that the effects of socialization and familial environments can alter the outcome of social attitudes.

Alford, Funk, and Hibbing (2005) are another classical twin design study that looks for genetic influence between different political factors. Namely, they explore the effects of genes and preference formation and their effect on party identification. Their findings establish a relationship between genetic influence on preferences but point out that genes play an extremely limited role in party identification.

Using the data from Eaves et al. (1999) classical twin study, Hatemi, et al., (2009) reevaluated the survey responses. To form his research, he sought to evaluate the effect of genetic variance between types of twins. He used the data to look for the effects of genetic variance on both political party identification and the intensity of the party tie the selected twin possessed. Hatemi et al. did not find a particularly strong relationship between genetic variance and party choice. However, they found a meaningful relationship between genetic variance and the intensity of one's party bond.

Our study seeks to replicate the work of Hatemi, et al. (2009). We will be looking for the effects of genetic variance on both party identification and the intensity of the party tie like their study. We will be using an entirely new data set for our work.

Working Theory and Hypotheses

People have opinions on a range of political issues, and they express their opinions by supporting political parties. This suggests that a person's political attitude on numerous subjects can be inferred from their choice of a political party. Nonetheless, it is also believed that a person's political preferences may be influenced by hereditary characteristics. In other words, a person's political opinions and ideologies may be influenced by genetic factors. Thus, it makes sense to assume that a person's relationship with political parties may also be influenced by genetic features. In other words, some genetic features may make a person more inclined to support a particular political party or set of beliefs, while other traits may make them less so. Given the circumstances, these hypotheses imply that political preferences are intricate and multifaceted, impacted by both environmental and hereditary factors and that they are a significant element in determining a person's political identity and conduct.

Knowing that a person's political preferences may be influenced by genetic characteristics, we can anticipate that genetic variance will have a considerable impact on a person's political identity as determined by a twin study. By concentrating on the twin study's unit of analysis, we may determine that the independent variable is genetic variance, and the dependent variable is a political identity. The political identities of identical twins, who share 100% of their genetic material, will be compared to those of fraternal twins, who typically only share 50% of their genetic makeup. We can gauge

how much genetic variation influences political identification by contrasting the political ideologies of these two groups.

H_{0PID}: Genetic variance between twins is not related to their choice of party identification.

H_{1PID}: Genetic variance between twins is related to their choice of party identification.

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